

Advancing Cybersecurity Excellence

A Comprehensive Strategy for Kenyan Organizations

Delivered to: Computer Society of Kenya

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Masterclass Overview

01

Advanced Data Protection
Strategies

02

Security Governance & Policy

03

Penetration Testing & Red
Teaming

04

Cloud Security & Risk
Management

05

Securing Endpoint Devices

06

Incident Response Team
Development

The Kenyan Landscape

- 📌 **DPA 2019:** The primary framework for data privacy in Kenya.
- 🏢 **ODPC:** Office of the Data Protection Commissioner oversight.
- 📈 **Growth:** Rapid digitization in banking, agriculture, and e-government.



Section 01

Advanced Data Protection

Strategies for the Modern Digital Age

Encryption: Three Pillars

At Rest

Protecting data on disks and databases using AES-256 standards.

In Transit

Securing data movement via TLS 1.3 and end-to-end encryption.

In Use

Confidential computing and Homomorphic encryption for active processing.

The Risk Equation

Advanced data protection aims to minimize the residual risk after controls are applied.

$$\text{Risk} = \frac{\text{Threat} \times \text{Vulnerability} \times \text{Asset Value}}{\text{Control Strength}}$$



Kenya DPA 2019: Key Duties

Requirement	Description	Action for IT
Registration	Mandatory for most controllers/processors.	File via ODPC Portal.
Data Protection Officer	Required for high-risk processing.	Appoint or Outsource.
Breach Notification	Notify ODPC within 72 hours of discovery.	Develop Response Plan.
DPIA	Impact assessment for high-risk projects.	Conduct before launch.

Data *Minimization*

72%

Organizations that practice strict data minimization reduce breach impact by up to 72%. Collect only what is necessary, for a specific purpose, and for a defined duration.

Section 02

Security Governance

Frameworks and Policy Development

Governance vs Management

- | | |
|--|--|
|  Setting strategic direction. |  Implementing controls. |
|  Accountability and compliance. |  Day-to-day operations. |
|  Oversight of performance. |  Technical configuration. |

Policy Hierarchy

Policies

High-level mandatory statements (The "What").

Standards

Specific requirements or tools (The "Must Use").

Guidelines

Best practices and suggestions (The "Should").

Strategic **Alignment**

Security governance ensures that IT security efforts directly support the business objectives of the organization.

- ✓ Risk-based approach.
- ✓ Resource optimization.
- ✓ Value delivery.



Risk Management Cycle



Adopting ISO 27001

ISO/IEC 27001 is the gold standard for Information Security Management Systems (ISMS).

- ★ Global recognition.
- 🛡️ Holistic protection.
- 🤝 Builds stakeholder trust.



Cultivating Awareness



Training is the most cost-effective control in any governance framework.

Section 03

Red Teaming & Testing

Testing Defenses Through Adversarial Emulation

Pentest vs. Red Teaming

Penetration Testing

Focused on finding as many vulnerabilities as possible. Scoped and systematic.

Red Teaming

Focused on testing detection and response. Goal-oriented (e.g., "Steal the CEO's data").

The Purple Synergy

- 🔥 **Red Team:** The Attackers.
- 🛡️ **Blue Team:** The Defenders.
- 🔧 **Purple Team:** Collaborative testing to improve detections instantly.



The Attack Lifecycle



AI Red Teaming

Prompt Injection

Tricking LLMs into revealing sensitive data.

Data Poisoning

Corrupting the training data of local models.

Jailbreaking

Bypassing the safety guardrails of AI agents.

Section 04

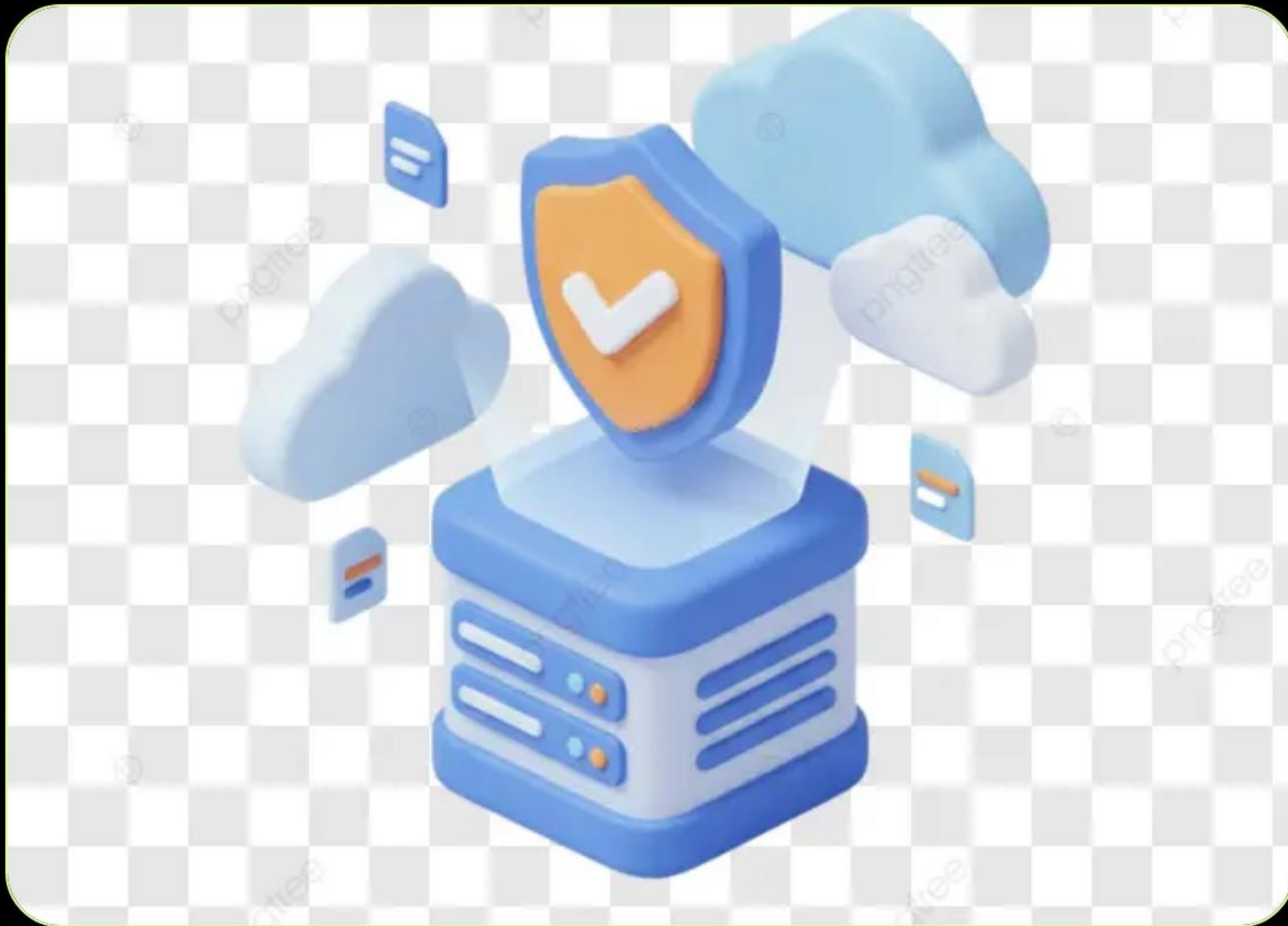
Cloud Security

Risk Management in the Cloud Era

Shared Responsibility

Layer	Provider Responsible	Customer Responsible
Infrastructure	Yes (Hardware/DC)	No
OS / Patching	IaaS (No), PaaS (Yes)	IaaS (Yes), PaaS (No)
Application	No	Yes
Data	No (Encryption tools only)	Yes (Ownership/Access)

Cloud Architecture



- 🔒 Multi-Factor Authentication (MFA).
- 🔧 Micro-segmentation.
- 🔍 Cloud Security Posture Management (CSPM).

Zero Trust Architecture



Eliminate the concept of a "trusted network." Every request must be authenticated, authorized, and continuously validated based on device health and user context.

Section 05

Endpoint Security

Protecting the Edge of Your Network

The Modern Endpoint

Workstations

Laptops & Desktops.

Mobile

BYOD Smartphones & Tablets.

IoT

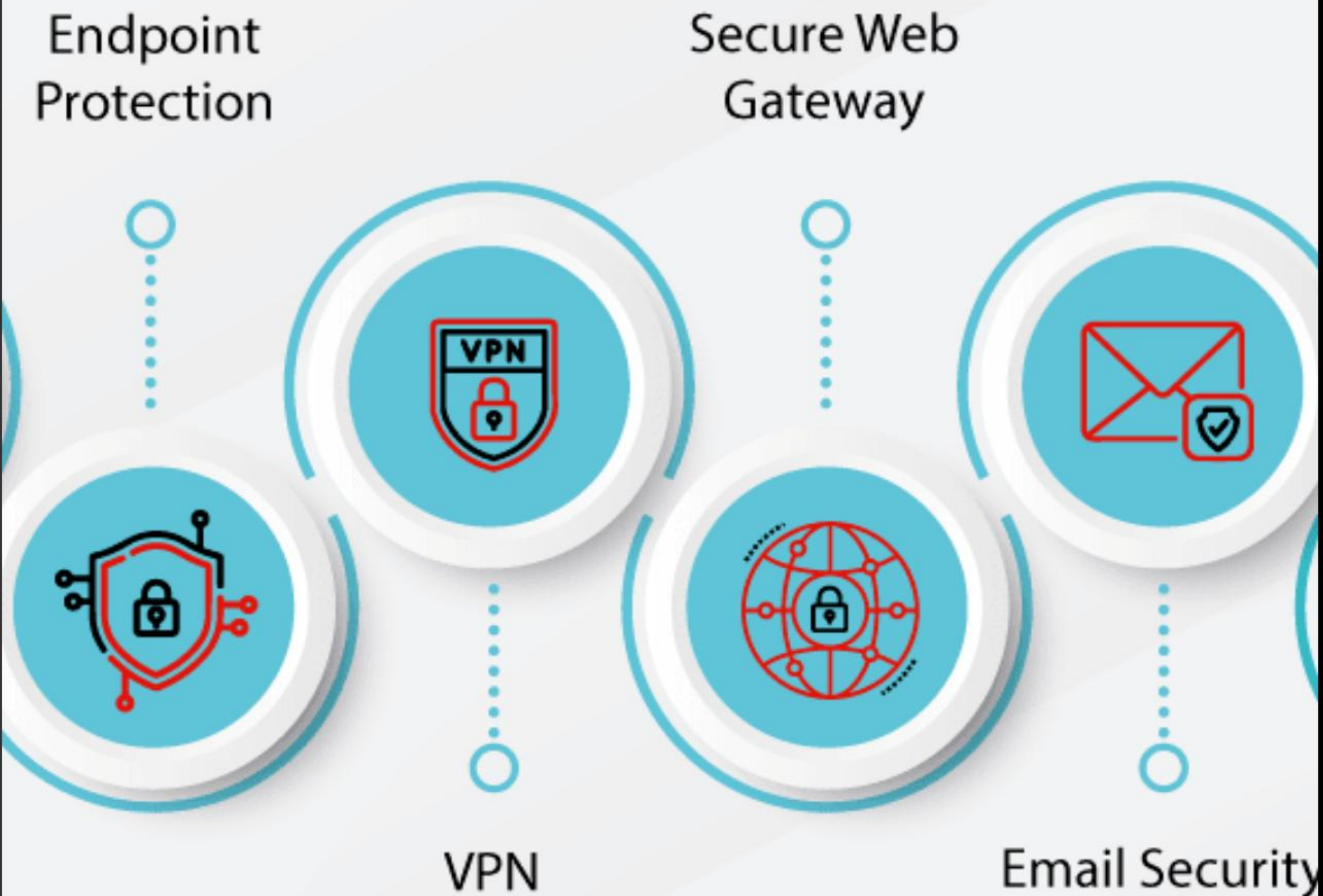
Smart sensors & Office devices.

EPP vs. EDR

Focuses on blocking known threats (AV, Firewall, HIPS).

Focuses on visibility, behavior analysis, and automated response.

Types of **Mobile Device Security**



Unified **Management**

Unified Endpoint Management (UEM) allows organizations to apply security policies across all devices from a single glass pane.

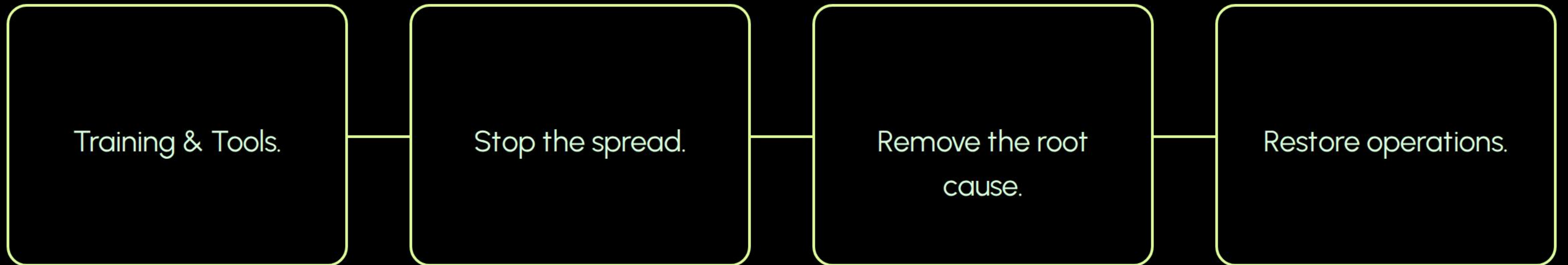
-  Remote Patching.
-  Remote Wipe.
-  Policy Compliance.

Section 06

Incident Response

Developing a World-Class Response Team

Incident Response Lifecycle



Building The CSIRT

Technical

Forensics, Network Analysts,
System Admins.

Strategic

Legal Counsel,
PR/Communications, HR.

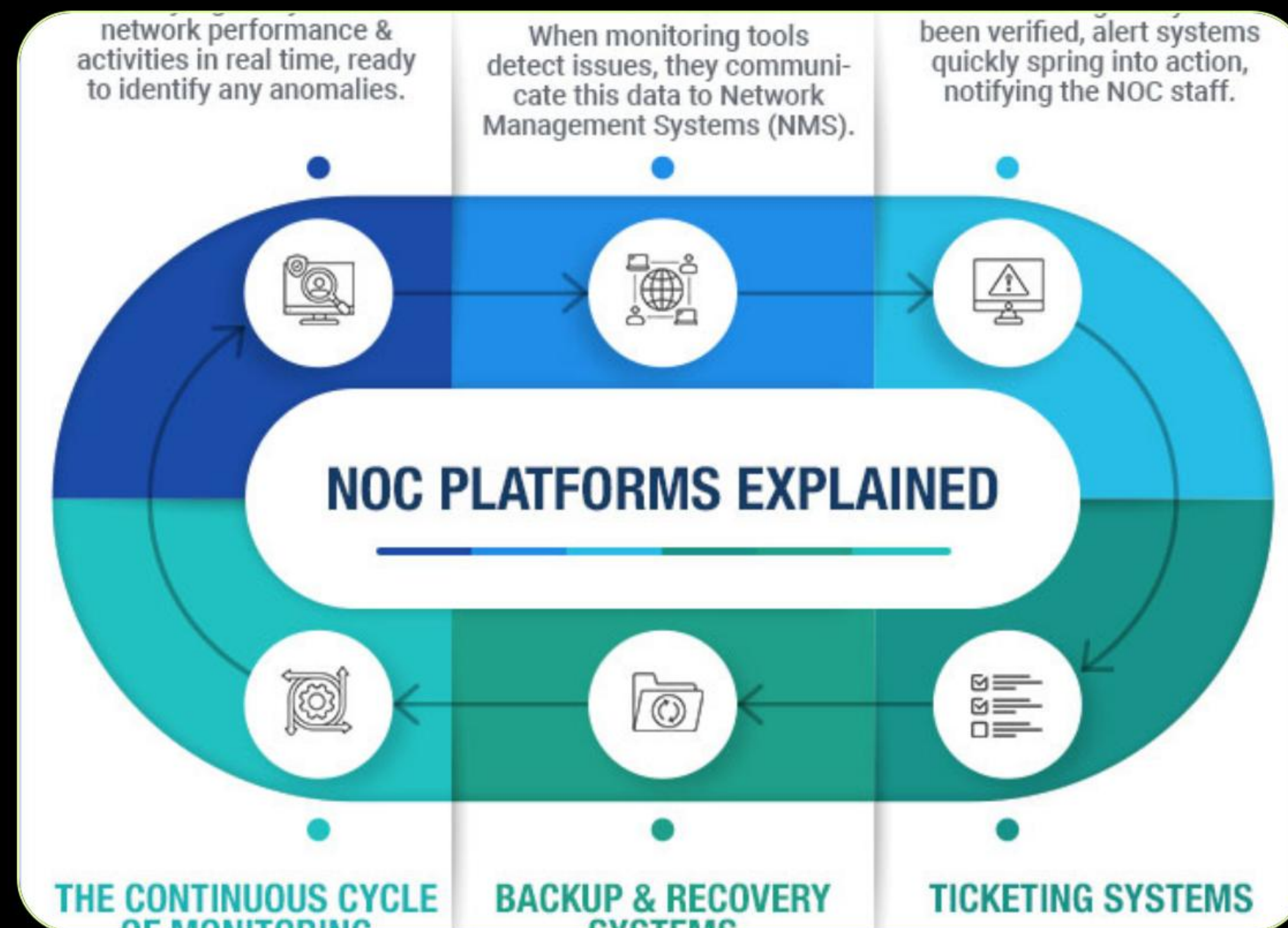
Leadership

CISO, Incident Commander.

The War Room

Efficient IR requires pre-defined playbooks for common scenarios like Ransomware or Phishing.

- 📄 Playbooks.
- 📡 Threat Intel.
- 💬 Communication Channels.



Questions & Answers

Thank you for your participation!

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